Urban Impervious Surface Automatic Threshold Detection Model Derived From Multitemporal Landsat Images

Yuliang Wang, *Member, IEEE*, and Mingshi Li

***Abstract*— Accurate mapping urban impervious surfaces (UIS) from satellite images is crucial for understanding urban land cover change directions and addressing UIS change-induced environmental issues. The UIS index is an effective way to reduce the computational cost of processing remote sensing imageries although continuous UIS detection from multitemporal Landsat images, repeated fine-tuning, and reselection for a proper threshold should be considered. In this article, we proposed a novel UIS index and an improved particle swarm optimization (PSO)-based UIS automatic threshold selection model (UISAT) to overcome those shortages when detecting UIS derived from multitemporal Landsat images. The proposed UIS index transformed different temporal Landsat images into index images with distinguishable pixel values for each gray pixel. In this way, the pixel gray values representing UIS were enhanced, and non-UIS was weakened with lower gray scales in the transformed index images. Next, an automatic threshold selection strategy was developed to enhance the applicability of the PSO model in terms of threshold fine-tuning and reselection. Then, we performed comprehensive assessments for the UISAT model, including in comparison to an improved convolutional deep belief network for relative accuracy verification, random points matching with Google Earth images, and comparing to other typical indices. The results demonstrated that the UISAT model provided the best performance in UIS detection from multitemporal Landsat images. The producer's accuracies (PAs) of UISAT results in Beijing, Hefei, and Guangzhou, China, were 91.73%, 93.87%, and 92.84%, while the user's accuracies (UA) were at 91.51%, 93.35%, and 92.24%. The proposed model outperformed the existing commonly used UIS indices [e.g., the Biophysical Composition Index (BCI), the Combinational Build-up Index (CBI), the Genetic Algorithm-based Urban Cluster Automatic Threshold (GA-UCAT), and the Gaussian-based Automatic Threshold Modified Normalized Difference Impervious Surface Index (G-MNDISI)] by an average increase of 6.51% in overall accuracy.**

***Index Terms*— Automatic threshold detection model, multitemporal Landsat images, urban impervious surfaces (UIS), urban index (UI).**

I. Introduction

URBAN impervious surfaces (UIS) are defined as anthropogenic materials that prevent water from penetrating directly into the ground and are considered as an increasingly important factor for urban planning and management [1]–[4]. Over the past two decades, as an important type of land cover, UIS has quickly sprawled globally. Urbanization is closely associated with land use/cover changes [5], [6]. In UIS regions, solar radiation is absorbed and reflected; air and heat fluxes of urban environments are weakened; and rainwater cannot participate in subsurface runoff [2], leading to several environmental issues, including urban heat and carbon island [7]–[10], nonpoint source water pollution [11], [12], and degraded air quality [13]. Accurate quantification of UIS, as well as an in-depth understanding of the spatial distribution, is critical to solving these environmental issues in built-up areas.

Remote sensing has the characteristics of multitemporal observation and large-scale imaging, thereby providing rich spatiotemporal information for UIS detection. Synthetic aperture radar (SAR) and Light Detection and Ranging (LiDAR) are active techniques to obtain ground images and discrete 3-D point clouds. They are not easily influenced by seasonal or environmental changes and allow for all day and night, all-weather, and high-resolution ground objects' observation [14]–[16]. However, fluctuating ground surface changes remain a major challenge for surveying via LiDAR and/or SAR techniques [17], [18]. High-spatial-resolution optical data (e.g., IKONOS, WorldView, and QuickBird) have been used for UIS mapping in local areas. These data are difficult to meet the need for quick mapping urban dynamics on a regional or global scale due to the processing complexity [19], [20]. Moreover, these high-resolution data contain many shadows and shade from high buildings or trees and so complicate UIS detection. Instead, coarse spatial resolution data (> 100 m), such as moderate-resolution imaging spectroradiometer (MODIS) datasets, have been applied in large-scale imaging with higher temporal resolution [21]. In particular, night light data have been used for mapping

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UIS expansion driven by people's activities, such as the Defense Meteorological Satellite Program/Operational Linescan System (DMSP/OLS) data and Luojia 1-01 nighttime light data [22], [23]. The night light data can help identify bright built-up areas from the dim non-UIS background. However, coarse spatial resolution data have lower UIS detection accuracy when mapping at local and regional scales [24], [25]. Compared with MODIS and night light data, Landsat data have better spatial resolution and long-time continuous imaging, making it more suitable for regional UIS mapping and change detection.

UIS mapping algorithms can be grouped into four categories in previous studies.

1) The spectral mixture analysis (SMA), which includes linear [26] and nonlinear SMA algorithms [27]. Medium-to-coarse resolution images have many mixed pixels, resulted from obvious spatial heterogeneity and fragmentation of ground objects. The linear SMA model aims for processing relatively simple homogeneous landscapes, while the nonlinear SMA model can process multitype landscapes. Appropriate endmember selection is a key step for a successful SMA, namely, choosing the number and type of endmembers and their corresponding spectral signatures [28]. Various methods to determine the number and type of endmembers in specific applied scenes have been described, including the V-I-S (vegetation, impervious surfaces, and bare soils) model and linear unmixing for urban scenes [29], [30]. SMA-based methods lead to a lack of detailed spatial information in medium-to-coarse spatial resolution remote sensing images [31].

2) Feature-based classification methods, including subpixel-, pixel-, and object-based methods. Typical methods include support vector machine (SVM) [32], random forest (RF) [33], and deep learning, such as the convolutional neural network (CNN) [34], [35], deep convolutional neural network (DCNN) [36], and convolutional deep belief network (CDBN) algorithms [37]. These methods can establish knowledge by learning prior knowledge and modify or update a knowledge base through self-learning of training samples and test samples; the approaches are also capable of completing self-recognition and environment recognition; and they are defined as supervised classification methods. Conversely, the unsupervised classification method (e.g., K-means and self-organizing maps) is seldom affected by human intervention, during the classification process, and possesses a lower operating cost [38].

3) Multisource data fusion models, such as spatial-based fusion, spatial-spectral-based fusion, and spatiotemporal-based fusion models [39], [40]. Multisource data fusion can take the respective advantages of different data sources and compensate for disadvantages in individual data sources. For example, optical remote sensing images combined with SAR or LiDAR data can overcome the problem caused by topographic relief and object height differences [41], [42]. Multisource data fusion can also obtain more valuable features, reduce the influence from "heterogeneous" and the imaging mechanisms of different sensors, and improve extraction accuracy for UIS detection. However, appropriate choices of data sources, fusion levels, and methods are keys to the success of multisource data fusion. Multisource data fusion models have three fusion levels: the pixel level, the feature level, and the decision level [2]. Each fusion level has specific limited conditions in practical applications. For example, optical and LiDAR or SAR data have been used for UIS mapping in a limited area due to the huge complexity of data [43].

However, SMA-based methods tend to overestimate dense buildings and underestimate spare buildings [2], while machine learning methods often have higher computation and time complexity (e.g., CNNs) compared with UIS index models. The classification accuracy of machine learning algorithms, such as SVM, depends closely on the quantity and quality of the samples. Spectral and spatial consistency issues, as well as the selection of a suitable fusion algorithm in multisource data fusion models, may seriously affect the extraction accuracy. Compared with the first three categories, UIS indices exhibit lower computational and temporal complexities, fewer operation condition constraints, and higher model execution efficiencies [44].

4) UIS indices, which use spectral combination operations to extract UIS information, such as the Combinational Build-up Index (CBI) [45] and the Biophysical Composition Index (BCI) [46]. In particular, UIS indices take UIS as the prominent foreground and non-UIS as the background to help to distinguish them. The UIS indices developed by previous studies can be categorized into the following three groups:

a) Normalized difference-based indices use the spectral characteristics of multispectral bands to calculate a new index image, and its pixel values are generally between –1 and 1 in the index images (gray-scale images of the results after the band operation). Specific examples include the Normalized Difference Built-up Index (NDBI) [47], the Normalized Difference Impervious Surface Index (NDISI) [48], the Modified Normalized Difference Impervious Surface Index (MNDISI) [49], the Normalized Difference Impervious Index (NDII) [50], the Urban Index (UI) [51], and the Enhanced Normalized Difference Impervious Surfaces Index (ENDISI) [52].

2) Feature extraction-based indices use the difference between the UIS and non-UIS types, such as the BCI, the Index-based Built-up Index (IBI) [53], the Built-up and Bare-land Index (BBI) [54], the Exponential Enhancing Impervious Surface method (p-index) [55], the Enhanced Built-up and Bareness Index (EBBI) [56], and the CBI.

3) Automatic threshold-based indices adopt a threshold algorithm to automatically select the optimal threshold. Typical representative models include the Genetic Algorithm-based Urban Cluster Automatic Threshold (GA-UCAT) [57] and the Gaussian-based Automatic Threshold MNDISI (G-MNDISI) [58].

Although these indices contributed to the development of UIS mapping and detection, several bottlenecks must be overcome. Few indices considered the effects of spatiotemporal differences in a single study. The spatiotemporal change affects the albedos of ground objects, resulting in reflectance variations of the same ground object. The thresholds of the indices must be readjusted to attain suitable extraction accuracy. In the aforementioned indices, some indices carried out subsidiary conditions during band operations. For example, BCI and NDBI mask water pixels before band operation. Moreover,

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TABLE I
DATASETS AND BANDS INFORMATION OF LANDSAT IMAGES SPANNING FOUR SEASONS IN THE THREE STUDY AREAS

Study areas	Beijing	Hefei	Guangzhou
Landsat Images	OLI: 15, Nov. 2017 (Winter)	OLI: 15, Jan. 2016 (Winter)	OLI: 7, Dec. 2016 (Winter)
	OLI: 4, Mar. 2017 (Spring)	TM: 19, Mar.2010(Spring)	OLI: 7, Feb. 2016 (Spring)
	TM: 26, Jul. 2011 (Summer) OLI: 19, Aug. 2014 (Autumn)	ETM+: 22, Jul.2012 (Summer) OLI: 11, Oct. 2015 (Autumn)	TM: 1, Jun. 2011 (Summer) OLI: 23, Oct. 2017 (Autumn)
Bands Selection and Wavelength (μm)	TM: Band 7: SWIR2 2.08-2.35 Band 4: NIR 0.76-0.90 Band 3: Red 0.63-0.69 Band 2: Green 0.52-0.60	ETM+: Band 7: SWIR2 2.09-2.35 Band 4: NIR 0.77-0.90 Band 3: Red 0.63-0.69 Band 2: Green 0.52-0.60	OLI: Band 7: SWIR2 2.11-2..29 Band 5: NIR 0.85-0.88 Band 4: Red 0.64-0.67 Band 3: Green 0.53-0.59

the majority of indices selected thresholds via artificial identification, which increased the workload and artificial interference and, consequently, reduced the accuracy of UIS detection from multitemporal images. Automatic threshold selection can overcome the drawback of manual threshold readjustment. GA-UCAT and G-MNDISI searched for a threshold to identify the foreground (UIS) pixels group from the background (non-UIS) pixels group. However, G-MNDISI and GA-UCAT only considered the influence of seasonal change and geospatial variation, respectively. The accuracy and dynamic adjustability need to be further improved for spatiotemporal change images in previous automatic threshold indices models.

In this article, we proposed a new UIS index (UISI) and a Particle Swarm Optimization (PSO)-based UIS Automatic Threshold selection model (UISAT) for UIS detection from multitemporal Landsat images. Our proposed model consists of three components; 1) a UISI; 2) an improved PSO-based automatic threshold selection algorithm; and 3) an integrated performance assessment strategy. The novel UISI employs multiband combinational operations to improve the UIS discrimination from non-UIS pixels in multispatiotemporal Landsat images. Furthermore, the threshold is automatically identified via an improved PSO algorithm to overcome the ground object's reflectance variation issue over time and space. The proposed model is tested with Landsat imagery collected considering three latitudinal and four seasonal gradients.

The rest of this article is organized as follows. The study area and datasets are presented in Section II, while the proposed UISI, UISAT, and performance assessment are described in Section III. The experiments and analysis of the proposed UISAT model are detailed in Section IV and discussed in Section V. Section VI concludes our work.

II. STUDY AREA AND DATASETS

A. Study Area

In recent decades, China has experienced faster urbanization in the world. Over the past four decades, the average growth ratio of urbanization was 3.1% in China compared with 0.75% in low-income countries and 1.65% in high-income countries. Particularly, the urbanization rate of China increased by 10.5% from 2010 to 2020 (the Development Research Center of the State Council, China, 2019). In this study, we chose three representative regions of China with apparent latitudinal gradients from north to south, namely, Beijing, Hefei, and Guangzhou, to test our proposed UIS detection model. Beijing is located in north China (39°24'–41°36' N, 115°42'–117°24' E), which belongs to a snow climate, with dry winter, and warm summer (Köppen classification: Dwb).

Beijing has a total area of 16 410.54 km^2, with the built area covers 1401 km^2. The population of permanent residents was given as 21.542 million, and the urbanization rate was 86.6% in 2019. Hefei is located in mid-east China (30°57′–32°32′ N, 116°41′–117°58′ E), with a warm temperate climate, fully humid, and hot summer (Köppen classification: Cfa). Hefei has a total area of 11 445.1 km^2 and a built-up area of 428 km^2. The population was 8.189 million, and the urbanization rate was 76.33% in 2019 [59]. Guangzhou is located in southern China (22°26′–23°56′ N, 112°57′–114°3′ E), with a total area of 7434 km^2. Guangzhou has a warm temperate climate and hot summer (Köppen classification: Cwa or Cfa) [60], with an average temperature of 20 °C–22 °C. The population was given as 14.4984 million, and the urbanization rate was 86.46% in 2019.

These three cities are the core cities of the Beijing-Tianjin-Hebei Region city cluster, the Yangtze River Delta city cluster, and the Pearl River Delta city cluster, respectively. They have experienced large-scale, rapid urban construction and development. The downtown areas of the three cities were selected for verifying the new index model (see Fig. 1).

B. Datasets

For each study area, we selected Landsat images over four seasons, covering three sensors in the three cities: Landsat 8 Operational Land Imager (OLI), Landsat 7 Enhanced Thematic Mapper Plus (ETM+), and Landsat 5 Thematic Mapper (TM), and they were downloaded from the United States (U.S.) Geological Survey (USGS) official website (https://glovis.usgs.gov/app). The Path/ Row tiles of the three study areas selected are Beijing (123/32), Hefei (121/38), and Guangzhou (122/44). The downloaded remotely sensed data were geometrically and atmospherically corrected to level 2 surface reflectance products by the USGS.

Detailed descriptions of the datasets are provided in Table I, and the reason for the band selection is described in Section III-A. We used eight OLI images, three TM images, and one ETM+ image. Different numbers of

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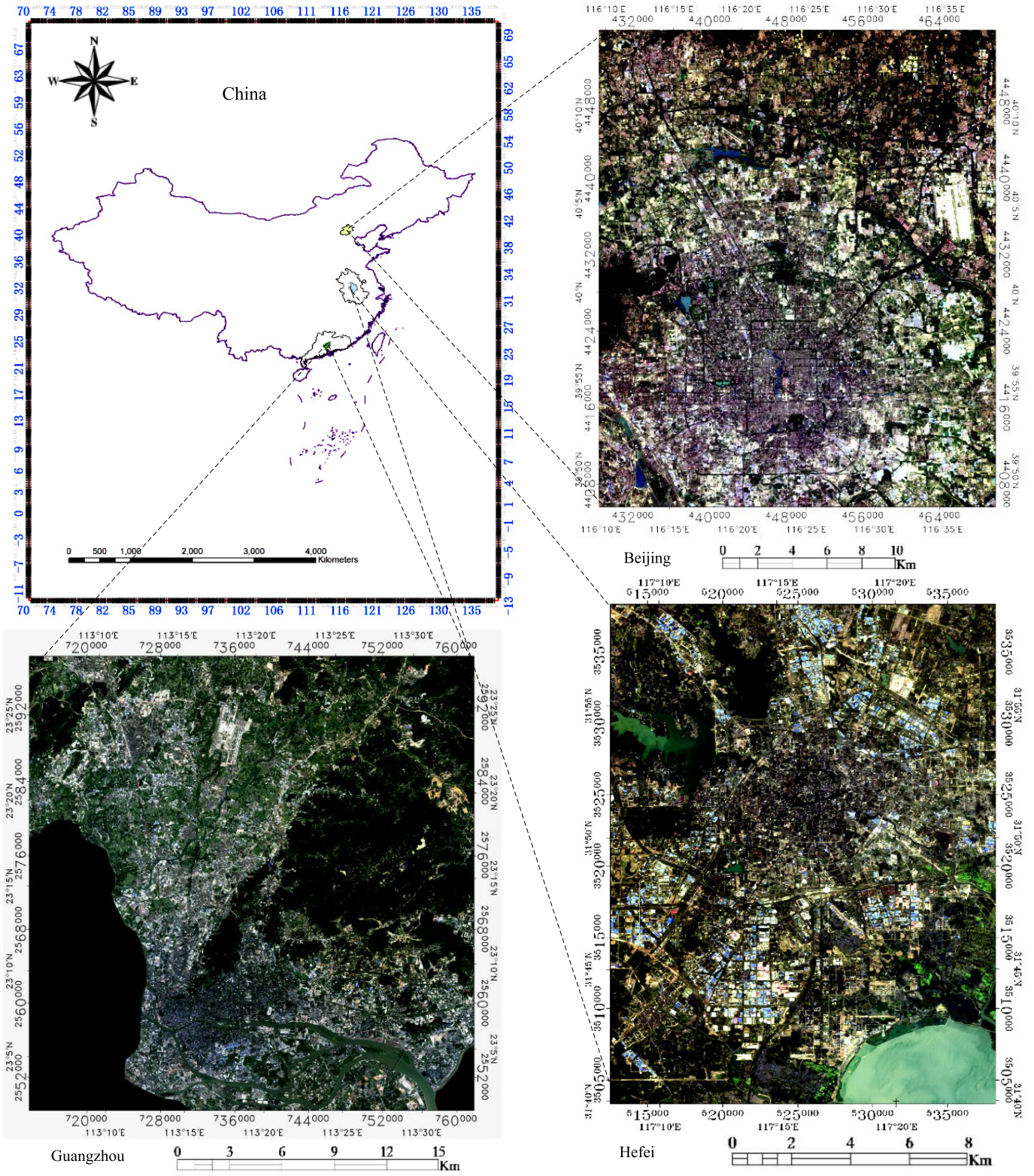


Fig. 1. Study area including three cities with apparent latitudinal gradients in China, including downtown areas of Beijing, Hefei, and Guangzhou running north to south.

images were used for each sensor type due to the image quality related to atmospheric disturbances. Cloud detection and declouding were applied for cloud-contaminated images by Fmask [61] and temporally adjacent pixels interpolation [62].

III. Methods

A. UIS Index (UISI)

The UIS index distinguishes UIS from non-UIS pixels according to the difference of pixel gray scale after spectral

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band calculations. Normalized difference UIS indices determine images with a gray value range between −1 and 1, with UIS pixels values greater than 0 and non-UIS pixels less than 0 [35]. However, this range is a function of the spectral band selection, with different spectral bands demonstrating different reflectivities for the same land cover. The land cover types can be divided into the following four classes from known ground objects in urban areas: 1) UIS (e.g., colored-roof buildings, business districts, airport, asphalt road, and old concrete buildings); 2) bare soil, including natural bare soil (uncultivated farmland) and artificial bare soil (construction site), as artificial bare soil has a higher albedo compared with natural bare soil; 3) vegetation (e.g., forest, grassland, urban green space, and farmland), and 4) water (e.g., lakes, river, and reservoirs). The classes (2)–(4) are non-UIS.

We compared the spectral reflectance of the same land cover region across different bands of multitemporal images (see Fig. 2). In Fig. 2, the same land cover has slightly different reflectivity cover four seasons in all bands. The reflectivity of UIS is almost higher than that of other classes in near-infrared (band) (NIR), short-wave (length) infrared (band) (SWIR)1, and SWIR2 bands. Bare soil reflectance is similar to that of UIS and can, thus, be confused with UIS in certain bands (e.g., blue and green bands). The reflectance of vegetation has a large fluctuation with seasonal transitions and obtains a higher value in summer for the NIR band than other land cover classes. The subtle spectral reflectance change of the four land cover classes can be seen over four different seasons in the same spectral band.

Fig. 2(a) presents the seasonal spectral reflectance of different land cover types in Beijing from OLI and TM imagery. The red vertical line segments denote the identifiable bands for UIS, with SWIR2 and SWIR1 identifying UIS and bare soil cover across the four seasons. The NIR band was able to identify UIS from vegetation and water during summer, while the UIS reflectance was higher than that of water and vegetation in the red and green bands.

Fig. 2(b) and (c) presents the seasonal spectral reflectance of different land cover types in Hefei and Guangzhou, respectively, from OLI, ETM+, and TM images. The reflectance of water and vegetation showed large fluctuations over the seasonal transition. The vegetation reflectance was highest in the summer in the NIR band due to the leaf structure of the vegetation. The spectral reflectance of all ground objects was observed to vary across seasons and locations. The spectral standard deviation of UIS was greater in winter compared to summer. This is attributed to the north–south latitudinal differences in the three cities.

The red vertical line segments in Fig. 2 reveal information that can be used to discriminate different land types from the multitemporal images. For the Landsat images, the identification of land cover is a function of season and band. However, SWIR1 and SWIR2 were able to detect land cover types for four seasons; in particular, UIS (high and low-albedo) pixels were identified from non-UIS pixels. The vegetation exhibited a wave crest in the NIR band and a trough in the red band. Furthermore, water demonstrated a wave crest in the green band and a trough in the SWIR1 and SWIR2. Combining the green and red bands allowed for the identification of UIS and vegetation, while the green and SWIR1 combination could identify water. Previous UISI models (e.g., UI, EBBI, and IBI) combined the NIR, SWIR1, SWIR2, red, and green bands as follows:

$$\mathrm{UI} = \frac{\mathrm{SWIR2} - \mathrm{NIR}}{\mathrm{SWIR2} + \mathrm{NIR}} \tag{1}$$

$$\mathrm{EBBI} = \frac{\mathrm{NIR} - \mathrm{Red}}{10\sqrt{\mathrm{NIR} + \mathrm{Red}}} \tag{2}$$

$$\mathrm{IBI} = \frac{\frac{2\mathrm{SWIR1}}{\mathrm{SWIR1}+\mathrm{NIR}} - \left(\frac{\mathrm{NIR}}{\mathrm{NIR}+\mathrm{Red}} + \frac{\mathrm{Green}}{\mathrm{Green}+\mathrm{SWIR1}}\right)}{\frac{2\mathrm{SWIR1}}{\mathrm{SWIR1}+\mathrm{NIR}} + \left(\frac{\mathrm{NIR}}{\mathrm{NIR}+\mathrm{Red}} + \frac{\mathrm{Green}}{\mathrm{Green}+\mathrm{SWIR1}}\right)}. \tag{3}$$

The vegetation reflectance peaked in the NIR band, while the SWIR1 and SWIR2 bands exhibited a stronger ability to identify UIS from other land cover types. Based on these observations, we constructed a new UISI using the SWIR2, NIR, red, and green bands as follows:

$$\mathrm{UISI} = \frac{5\varphi \times (\mathrm{SWIR2} + \mathrm{Red\text{-}NIR\text{-}Green})}{\mathrm{SWIR2} + \mathrm{Red} + \mathrm{NIR} + \mathrm{Green}} - \varphi \times \lambda \tag{4}$$

$$\text{s.t. } \lambda = \frac{\mathrm{Green} - \mathrm{SWIR2}}{\mathrm{Green} + \mathrm{SWIR2}} \tag{5}$$

where φ is an adjustment parameter (empirical value) that is used to improve the readability of the pixel gray values in index images. The value of φ can be determined according to the requirement on given spatiotemporal Landsat images. In this experiment, φ ranged from 1 to 3. The experiment showed that the pixels' gray values contrast increased between UIS and non-UIS with large values not easy to calculate when $\varphi > 3$ and had a poor gray values contrast when $\varphi < 1$. In this article, we selected $\varphi = 2$ (the mean of 1 and 3) for the UISI model. The SWIR2, red, NIR, and green results demonstrated surface reflectance corresponding to the bands. λ could improve the identification of water, lowering the pixels' gray values after index calculation. λ had a similar effect, such as the modified normalized difference water index (MNDWI) [63] and reduced pixels gray values of water in the index image. The NIR band weakened the effect of bare soil, and the green band weakened the pixel gray values of water in the formula numerator.

Bare soil is the principal disturbance factor for interfering with the extraction of UIS. The proposed UISI model could enhance the pixel gray values difference of land cover types. The gray values of vegetation, water, and bare soil pixels were weakened in the index image. We will discuss this performance in Section IV.

The UIS extraction by using the proposed model can be divided into two steps: 1) spectral band math and 2) the threshold selection for the segmentation of UIS and non-UIS. In particular, we determined a threshold for high accuracy in UIS extraction from the index image. The pixel gray value was divided into m gray levels in terms of the gray value range of the index image, with the gray level dataset defined as $T_{\mathrm{UISI}} = \{T_0, T_1, T_2, \ldots, T_k, \ldots, T_n\}(0 \le n \le m, 0 \le k \le n)$, and we assumed T_k as the best threshold; then, the UIS threshold segmentation condition should be expressed as follows:

$$\mathrm{UISI}_T = \begin{cases} \mathrm{UIS}, & T_i \ge T_k \\ \mathrm{non-UIS}, & T_i < T_k \end{cases} 0 \le i \le n. \tag{6}$$

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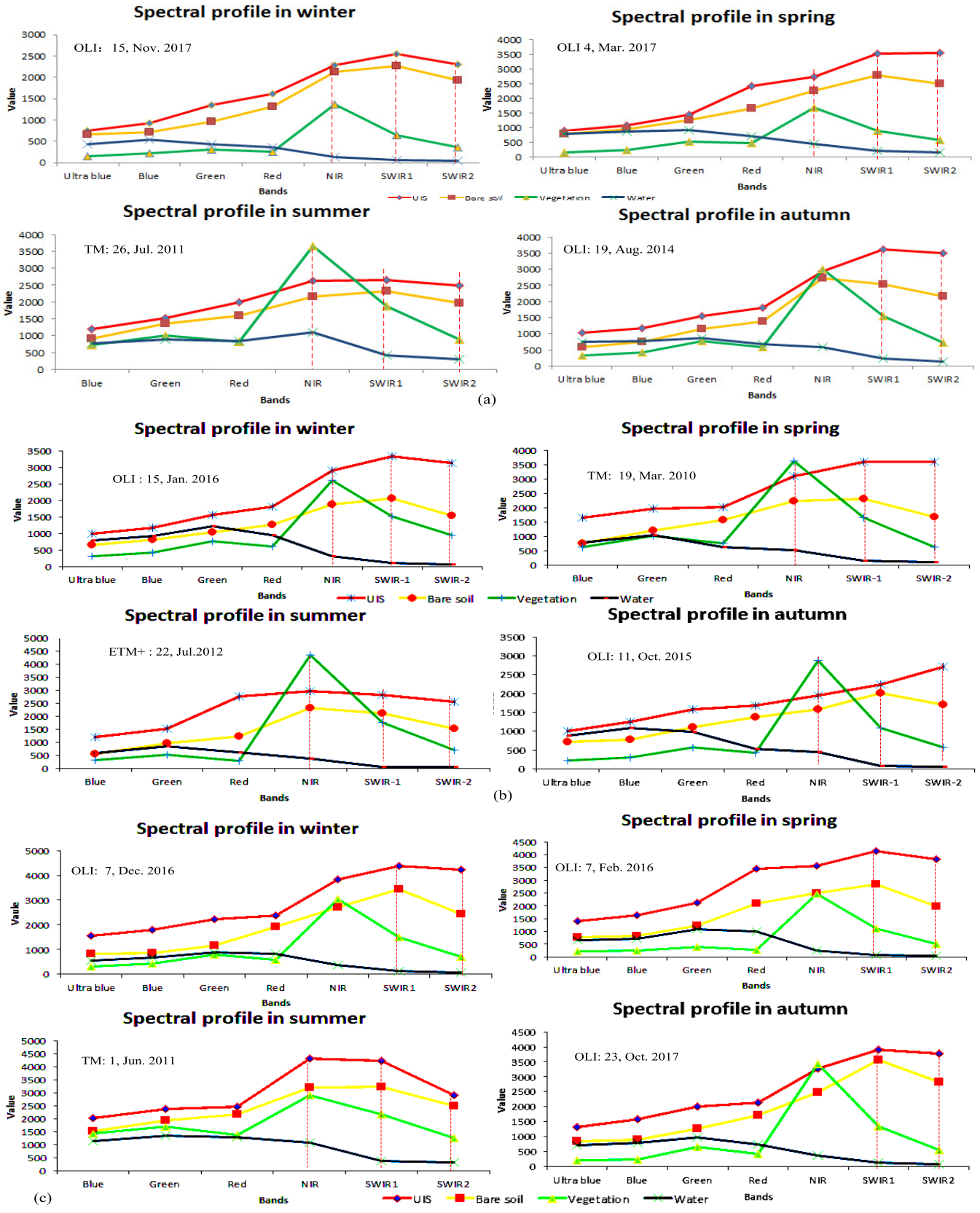


Fig. 2. Surface reflectance comparison of known ground object classes in multitemporal Landsat images. (a)–(c) Reflectance of the same ground object classes covering four seasons in Beijing, Hefei, and Guangzhou, China, respectively. The red vertical line segments mark identifiable bands.

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where UISI_T represents the pixels gray-scale group as determined from the threshold segmentation. However, the threshold was fine-tuned due to differences in the multitemporal Landsat images, and an automatic threshold selection algorithm was implemented to avoid repetitive manual steps.

B. Improved PSO-Based Automatic Threshold Algorithm

The automatic threshold method can determine the optimal segmentation threshold and overcome the spectral differences resulted from multitemporal Landsat images. In this article, the PSO-based method was employed to identify an optimal threshold. The PSO-based method is an evolutionary computation that iteratively updates the solution. It improves the optimization quality in the global candidate solution space via cooperation and information sharing among individuals within the group [64], [65]. The method is composed of four key steps.

1) *Particle Swarm Initialization:* This defines the swarm scale, initial position, the speed of the particle, and other relevant parameters.

2) *Fitness Calculation for Each Particle:* This initializes the individual and global optimal temporary position and obtains the individual and global temporary fitness value.

3) *Fitness Comparison:* This compares the fitness between individual particles, individual optimal particles, and global optimal particles and updates the individual and global optimal particles.

4) *Updating:* This step updates the position and speed of the particle. Step 2) is executed iteratively until the globally optimal particle is determined.

In an index image, the gray level is divided into m groups and for every group with an initial segmentation threshold; their dataset can be expressed as $T_{\mathrm{UISI}} = \{T_0, T_1, T_2, \ldots, T_k, T_i \cdots T_n\}(0 \le n \le m, 0 \le k \le n)$, $i(0 \le i \le n)$ can be as a particle of the T_i group. Every particle has a motor velocity defined as V_i, and the vector set of velocities is $V_{Ti} = \{V_{i0}, V_{i1}, V_{i2}, \ldots, V_{\mathrm{in}}\}$. The initial optimal threshold is defined as T_{bi} for the ith particle, and the global initial optimal threshold is defined as T_{gi}. T_{bi} and T_{gi} are solutions to the fitness function.

In this study, the image entropy was applied for the fitness function of PSO to improve the threshold selection. The image entropy can represent the aggregation feature of the image gray-scale distribution, and the function is presented as follows:

$$F(T_k) = H(T_k) = H_0(T_k) + H_1(T_k) \tag{7}$$

where $F(T_k)$ is the fitness function of PSO in threshold T_k, $H(T_k)$ is the total of the image entropy, and T_k is the current optimal threshold, which divides index images into the background ($< T_k$) and foreground ($\ge T_k$). $H_0(T_k)$ and $H_1(T_k)$ represent the image entropy of background and foreground in index images, respectively.

The calculation procedures are presented by the following functions:

$$H_0(T_k) = -\sum_{i=0}^{T_{k-1}} \frac{P(i)}{P_0(T_k)} \log\left(\frac{P(i)}{P_0(T_k)}\right) \tag{8}$$

$$H_1(T_k) = -\sum_{i=T_k}^{n-1} \frac{P(i)}{P_1(T_k)} \log\left(\frac{P(i)}{P_1(T_k)}\right) \tag{9}$$

where $P_0(T_k)$ and $P_1(T_k)$ are the cumulative probabilities of each gray level for the background and foreground groups, respectively. The relationship between $P_1(T_k)$ and $P_0(T_k)$ can be expressed as follows:

$$P_1(T_k) = 1 - P_0(T_k) \tag{10}$$

$$P_0(T_k) = \sum_{i=0}^{T_{k-1}} p(i). \tag{11}$$

The probability of each gray level in the background groups is

$$\frac{p(0)}{p_0(T_k)}, \frac{p(1)}{p_0(T_k)}, \frac{p(2)}{p_0(k)}, \ldots\ldots, \frac{p(k-1)}{p_0(T_k)} \tag{12}$$

and the probability of each gray level in foreground groups is

$$\frac{p(k)}{p_1(T_k)}, \frac{p(k+1)}{p_1(T_k)}, \frac{p(k+2)}{p_1(T_k)}, \ldots\ldots, \frac{p(n-1)}{p_1(T_k)}. \tag{13}$$

The purpose of the PSO-based method is to determine the global optimal value as the threshold via the fitness function. This method updates the individual and global optimal particles through fitness value comparisons and iteratively searches this process until the best global threshold is found. The position and speed of the particle update function are expressed as follows:

$$T_i = T_i + V_i \tag{14}$$

$$V_i = \omega * V_i + \beta_1 * \theta * (T_{bi} - T_i) + \beta_2 * \theta * (T_{gi} - T_i) \tag{15}$$

where β_1 and β_2 are the learning factors (constant 2), θ is a random number between 0 and 1, and T_{bi} and T_{gi} are defined as the initial optimal threshold and the global initial optimal threshold; they are nonzero. The value of ω is the inertia factor ($\omega \ge 0$). The inertia factor ω can adjust the global optimization ability of the algorithm and adopt the linearly decreasing weight (LDW) [66], [67], which is presented as follows:

$$\omega = \omega_{\mathrm{LDW}} = (\omega_{\max} - \omega_{\min}) * \frac{(IT_{\max} - IT_{\mathrm{cur}})}{IT_{\max}} + \omega_{\min} \tag{16}$$

where $\omega_{\max}$ is the maximum inertia weight of 0.9 and $\omega_{\min}$ is the initial inertia weight of 0.4. $IT_{\max}$ and IT_{cur} represent the maximum number of iterations and the current number of iterations, respectively. The dynamic weight ω allows the PSO-based method to have a better global convergence ability in the early stage and better local convergence ability in the later stage.

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C. Performance Verification and Evaluation

The UIS detection accuracy of the proposed method was verified using an improved CDBN (dc-CDBN) for relative accuracy comparisons [37]. The relative accuracy was found using an accuracy verification approach where a more accurate algorithm is taken as the comparison object [56]. Specifically, we counted the number of UIS and non-UIS pixels from the dc-CDBN results and index images in the same regions. The relative accuracy of the index model was obtained according to the proportion of UIS and non-UIS pixels in the dc-CDBN results. The dc-CDBN is an unsupervised learning method that can construct and train a sample space by self-learning, thus improving the automation and intelligence of UIS detection. The performance of the proposed model was compared with typical UIS indices (see Section IV for more details). We randomly selected several areas for random point verification via the comparison of ground objects from Google Earth imagery (2.39 m spatial resolution) and ground survey data [67].

The pixels were divided into either foreground (UIS) or background (non-UIS) gray-level groups using the UISI index image threshold. The performance assessment of the threshold segmentation was based on the interclass correlation coefficient (ICC) algorithm [68]. ICC is a statistical method that measures the reliability and consistency of the pixel gray level with values ranging between 0 and 1 and higher values representing a better threshold segmentation. The pixels gray-level groups of the background and foreground from a UISI index image were, respectively, defined as $U_i^B = \{x_i\}_{i=0}^{k-1}$ and $U_i^F = \{x_i\}_{i=k}^{m-1}$, where k is the position of a gray pixel at an optimal threshold T_k, x_j is the gray pixel of the neighboring pixel, $i \neq j$, and $i, j = 0, 1, \ldots, k, \ldots, m-1$. m is the gray levels in terms of the gray value range of the index image. Taking the foreground gray pixel group as an example, the ICC of the foreground gray pixel group $\mathbb{R}_F$ can be expressed as follows:

$$\mathbb{R}_F = \frac{\sum_{i,j=k}^{m-1}(x_i - \bar{X})(x_j - \bar{X})}{(m-1)\sigma_s^2}, \quad \mathbb{R}_F \in [0, 1] \tag{17}$$

where

$$\bar{x} = \frac{\sum_{i,j=k}^{m-1}(x_i + x_j)}{2(m-1)}, \quad \sigma_s^2 = \frac{\sum_{i,j=k}^{m-1}(x_i - \bar{x})^2 + (x_j - \bar{x})^2}{2(m-1)}. \tag{18}$$

The ICC of the background pixels group $\mathbb{R}_B$ was calculated in the same way. The ICC can reflect the effect of threshold segmentation with higher $\mathbb{R}_F$ values denoting a better threshold for the UISI index images.

D. UISAT Model

The UISAT model provides an effective solution for UIS mapping and detection from multitemporal Landsat images. The proposed UISI model can effectively enhance the visualization of the built-up areas and suppress the non-UIS (e.g., bare soil, vegetation, and water) displays in index images. To reduce the influence of multitemporal Landsat images on threshold fine-tuning, we employed an automatic threshold selection using the improved PSO-based method. Fig. 3 presents the operational processes and relationships of the three modules within the model, namely, the mapping and detection of UIS, the algorithm optimization process, and the verification and evaluation.

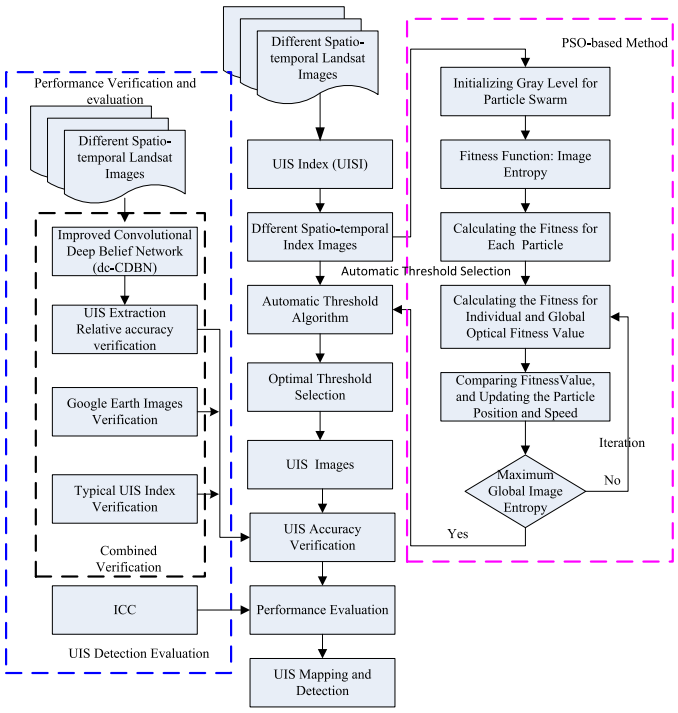


Fig. 3. Flowchart of the UISAT model. The pink rectangle is an automatic threshold selection method, and the blue rectangle is the performance verification and evaluation of the model.

IV. Experiments and Results

A. Performance of UISI

The performance of the UISI model was verified using multitemporal Landsat images (OLI, ETM+, and TM). We selected $\varphi = 2$ for the UISI calculations, and Landsat images over four seasons in the three study areas were translated into UISI index images (see Fig. 4). The gray values of UIS varied with the season within the same urban area. The visual contrasts of various land cover types were distinct in the spring and summer due to the vegetation cover. For example, there was a higher gray difference between UIS and non-UIS in summer in Hefei. The regions of water and vegetation appeared with low gray values in the index images, while bare soil and UIS were gray and light gray, respectively. The identification degree between UIS and bare soil significantly improved compared to the original images. Vegetation coverage can reduce the disturbance of bare soil, particularly during the late spring, summer, and early autumn in mid-northern China (Beijing and Hefei) and southern China (Guangzhou). Moreover, the gray value of the same ground objects was different in the index images derived from different mapping sensors, such as ETM+ image (see Fig. 4(b). There had been significant discrimination of gray value between UIS and non-UIS.

Bare soil disturbance is a key factor influencing the effectiveness of the UISI model. The spectral reflectance and index value of bare soil and UIS regions are compared in Fig. 5.

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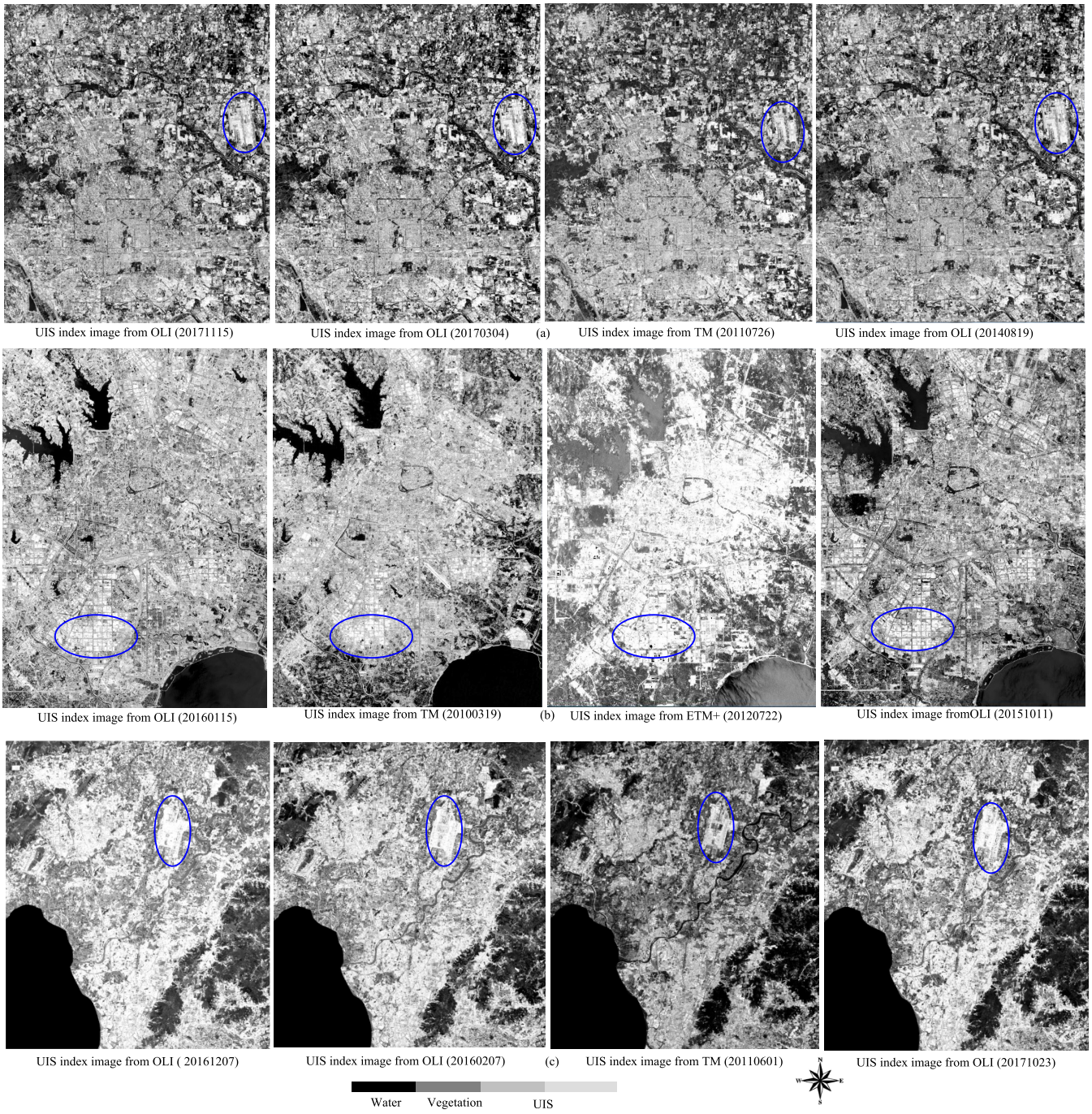


Fig. 4. Pixels gray values change in index images from multitemporal Landsat images. (a)–(c) UISI gray-scale images in the downtown of Beijing, Hefei, and Guangzhou, China, respectively. Higher brightness surfaces are UIS and lower brightness surfaces are non-UIS (water, vegetation, and bare soil). The ellipses mark pixels' gray change area from multitemporal Landsat images.

Fig. 5(d)–(l) compares the bare soil attenuation between the proposed UISI model and typical indices (NDII, NDBI, IBI, UI, EBBI, BCI, and CBI) in Hefei from Landsat OLI (October 11, 2015). The horizontal axis is the index value, and the vertical axis is the frequency of occurrence of the index value in the index image. Fig. 5(a) presents the Landsat OLI image with the green and red ellipse boxes representing the bare soil and UIS, respectively, as a comparison sample. Fig. 5(b) demonstrates the UISI image with corresponding bare soil and UIS regions. Fig. 5(c) depicts the spectral confusion overlap between the bare soil and UIS in the Landsat image. The blue rectangular boxes denote the confusion or overlapping areas of the bare soil and UIS histograms. The confusion areas of the index image exhibit a marked decrease compared to the original image. The UISI model obtained the minimum confusion area amongst all indices.

To further demonstrate the performances of the UISI, we demonstrate the UISI index derived for the Beijing

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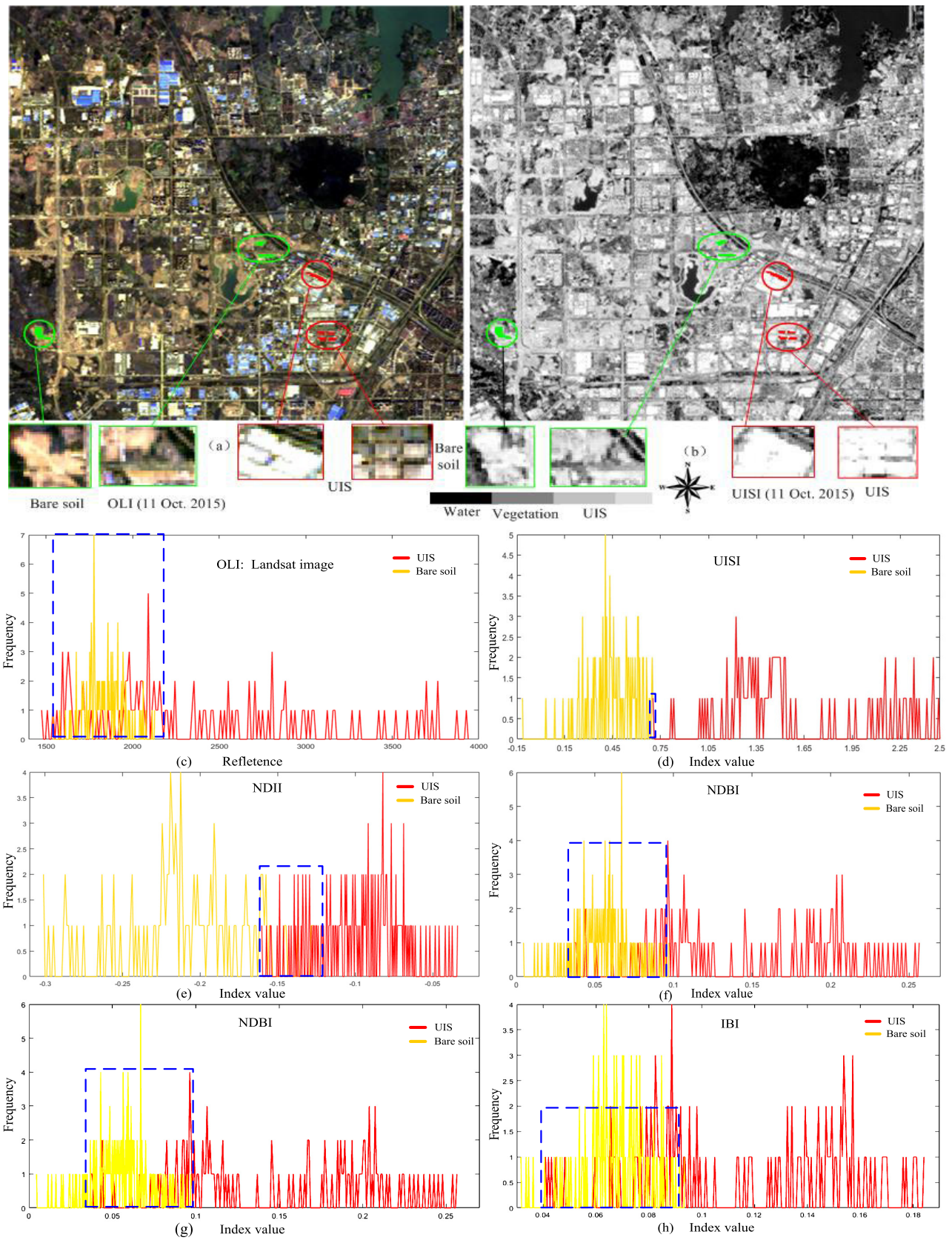


Fig. 5. Comparison of the confusion degree between UIS and bare soil. Green and red ellipse boxes are bare soil and UIS in (a) and (b), respectively, and blue rectangular boxes show the spectral confusion or index value overlapping areas between bare soil and UIS. (a) Landsat image (combination of band 4-R, 3-G, and 2-B) in Hefei, China. (b) Corresponding index image derived from (1). (c)–(l) Spectral confusion or index value overlapping histograms of bare soil and UIS derived from original, UISI index images, and other typical indices images, respectively. The horizontal axis is the index value, and the vertical axis is the frequency of index value occurrence in the index images.

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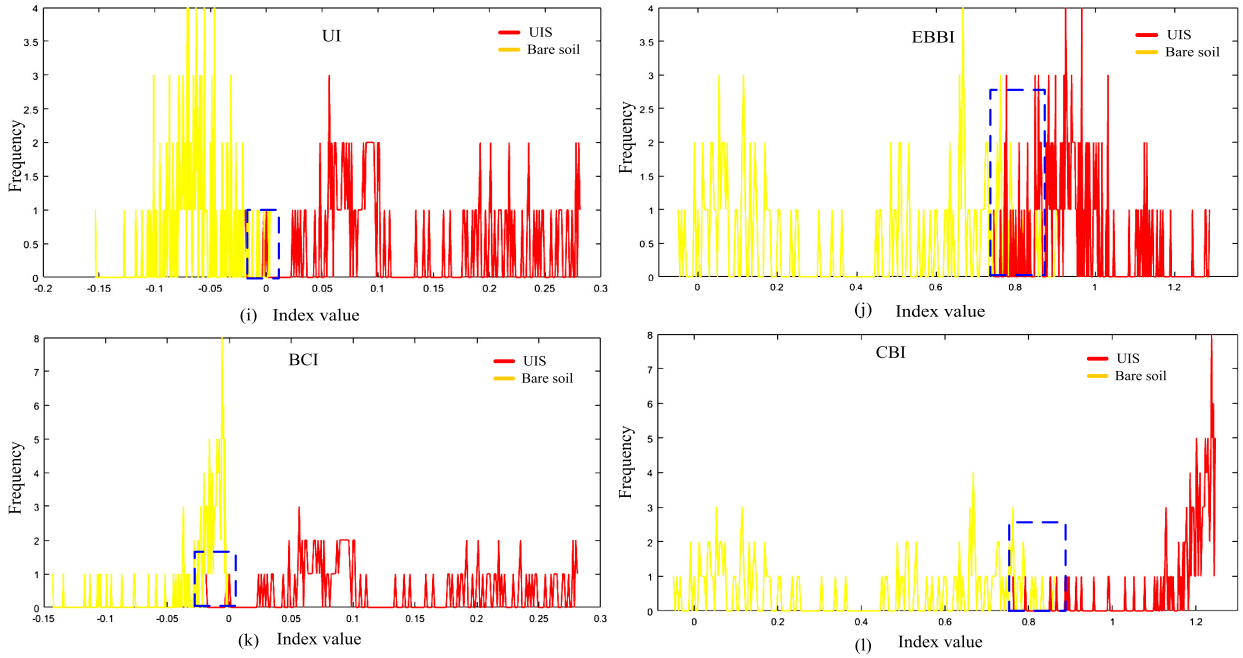


Fig. 5. *(Continued.)* Comparison of the confusion degree between UIS and bare soil. Green and red ellipse boxes are bare soil and UIS in (a) and (b), respectively, and blue rectangular boxes show the spectral confusion or index value overlapping areas between bare soil and UIS. (a) Landsat image (combination of band 4-R, 3-G, and 2-B) in Hefei, China. (b) Corresponding index image derived from (1). (c)–(l) Spectral confusion or index value overlapping histograms of bare soil and UIS derived from original, UISI index images, and other typical indices images, respectively. The horizontal axis is the index value, and the vertical axis is the frequency of index value occurrence in the index images.

TABLE II

AVERAGE INDEX VALUE OF FOUR CLASSES OF LAND COVERS IN UISI INDEX IMAGES DERIVED FROM MULTITEMPORAL LANDSAT IMAGES IN BEIJING, HEFEI, AND GUANGZHOU, CHINA

City	The classes of land cover	Winter	Spring	Summer	Autumn
Beijing	UIS	-0.22	-0.128	0.016	-0.0775
	Bare soil	-1.722	-1.726	-1.513	-1.535
	Vegetation	-5.113	-3.341	-3.146	-5.063
	Water	-4.261	-3.921	-4.715	-4.128
Hefei	UIS	1.0375	1.276	1.684	1.091
	Bare soil	-0.516	-0.926	-0.627	-0.893
	Vegetation	-4.144	-4.131	-3.363	-3.182
	Water	-3.708	-4.734	-4.432	-3.354
Guangzhou	UIS	0.8505	0.8645	0.6085	0.84
	Bare soil	-1.166	-0.75	-1.977	-1.129
	Vegetation	-4.917	-5.433	-4.313	-5.737
	Water	-3.284	-3.159	-3.932	-4.816

Capital International Airport, Hefei Dashu Mountain, and Guangzhou Baiyun International Airport (see Fig. 6). Roads and buildings exhibited greater brightness values with clear structural information, while vegetation and water were associated with lower brightness values. The gray values within the UISI image were distinct for the UIS and non-UIS regions. Fig. 6 demonstrates that the proposed UISI model could translate the original multitemporal Landsat image into ground objects identifiable in the index images.

The pixel index value of UIS and non-UIS varied with the spatiotemporal changes of the UISI index images. In particular, ground objects exhibited a lower gray value in Beijing compared to Hefei and Guangzhou (see Fig. 4(a) and Table II). Such differences in ground object gray values were a function of multiple factors, including the sensor type, scanning and imaging time, density of building, and differences in regional ground object types.

Table II shows the average index values of four land cover classes derived from three areas and four seasons within the

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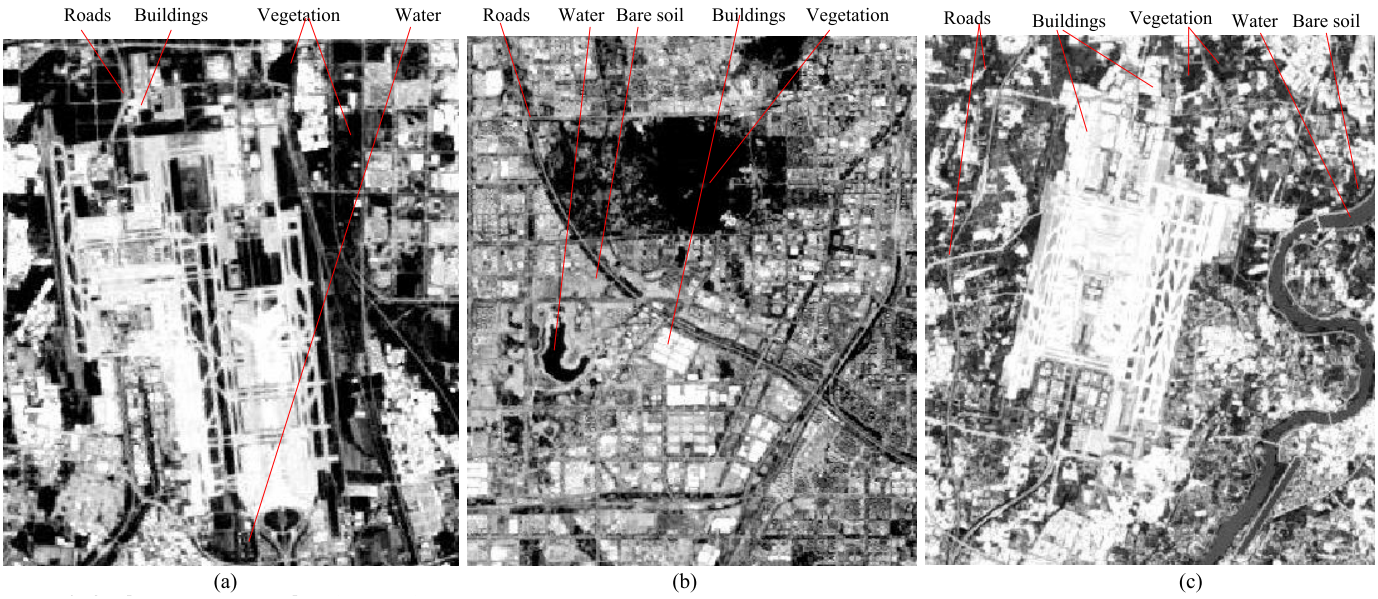


Fig. 6. Local magnification of three areas in UIS index images. (a) Beijing Capital International Airport. (b) Hefei Dashu Mountain areas. (c) Guangzhou Baiyun International Airport.

UISI images. The average index values of the four land covers in the different regions differed. For example, the average index value of UIS was around 0 in Beijing for four seasons, while the value was around 1.2 and 0.8 in Hefei and Guangzhou, respectively. The building density affected the UIS index value in UISI index images. The index value of UIS in the area with high building density was lower than that in the area with low building density. However, the average index value of four ground object types had an apparent identifiable value range in a single index image. The average index value of UIS (e.g., asphalt roads and old concrete buildings) was greater than other land cover types. Vegetation and water exhibited lower negative gray values, ranging from -3 to -5, while the average index value of bare soil was approximately between -0.5 and -2.

Certain bare soil and UIS regions demonstrated similar values; the index values of the latter were always greater than bare soil. In addition, the degree of bare soil disturbance varied with the spatiotemporal changes due to vegetation cover. In northern China, the main disturbance originated from uncultivated farmlands in the late autumn, winter, and early spring, while they were covered with vegetation in the late spring, summer, and early autumn. In comparison, except for a few artificial bare soil regions on construction sites, most of the natural soil in southern China was covered by vegetation. Section V will discuss this further.

B. Automatic Threshold Selection

The automatic threshold algorithm solved the problem of continuous UIS extraction from multitemporal Landsat images. The automatic threshold algorithm can obtain an optimal threshold by gray-level segmentation based on the distribution of gray values in the index images and can overcome the subjectivity of artificial thresholding. The improved PSO-based method employed the image entropy as a fitness function, with the solution of the maximum image entropy selected as an optimal threshold. At the initialization stage of the algorithm, the maximum and minimum gray values were extracted from the UISI image and taken as the range of positions in which the particle could move. The initial position of the particle and the total image entropy currently at the initial threshold T_0 were defined, followed by the calculation of the probability and probability sum of each gray level for the current threshold T_0. Learning factors β_1 and β_2 were set as 2, random number θ ranged from 0 to 1, and the inertia factor ω decreased as the number of iterations increased [66]. The optimal threshold T_k was selected to establish an appropriate gray segmentation between UIS and non-UIS regions in the index images.

In the experiments, the selection of the optimal thresholds was fine-tuned to consider the spatiotemporal variations in index images. The difference of the thresholds varied with the spatiotemporal change images. In Fig. 7, a comparison of threshold selection for multitemporal index images derived from different spatiotemporal Landsat images (OLI, ETM+, and TM) was demonstrated. If a pixel value of pixels was greater than or equal to the threshold T_k, then the pixel was recognized as UIS pixels; otherwise, it was classified into non-UIS types.

Fig. 7(a) and (b) shows the UIS threshold selection results for the index images derived from OLI and TM images in Beijing, China, where the optimal threshold T_k was close to -1. Fig. 7(c) and (d) shows the optimal thresholds that were greater than zero for Hefei, China, derived from ETM+ and TM images. Fig. 7(e) and (f) presents optimal threshold selection of the threshold from OLI and TM images for

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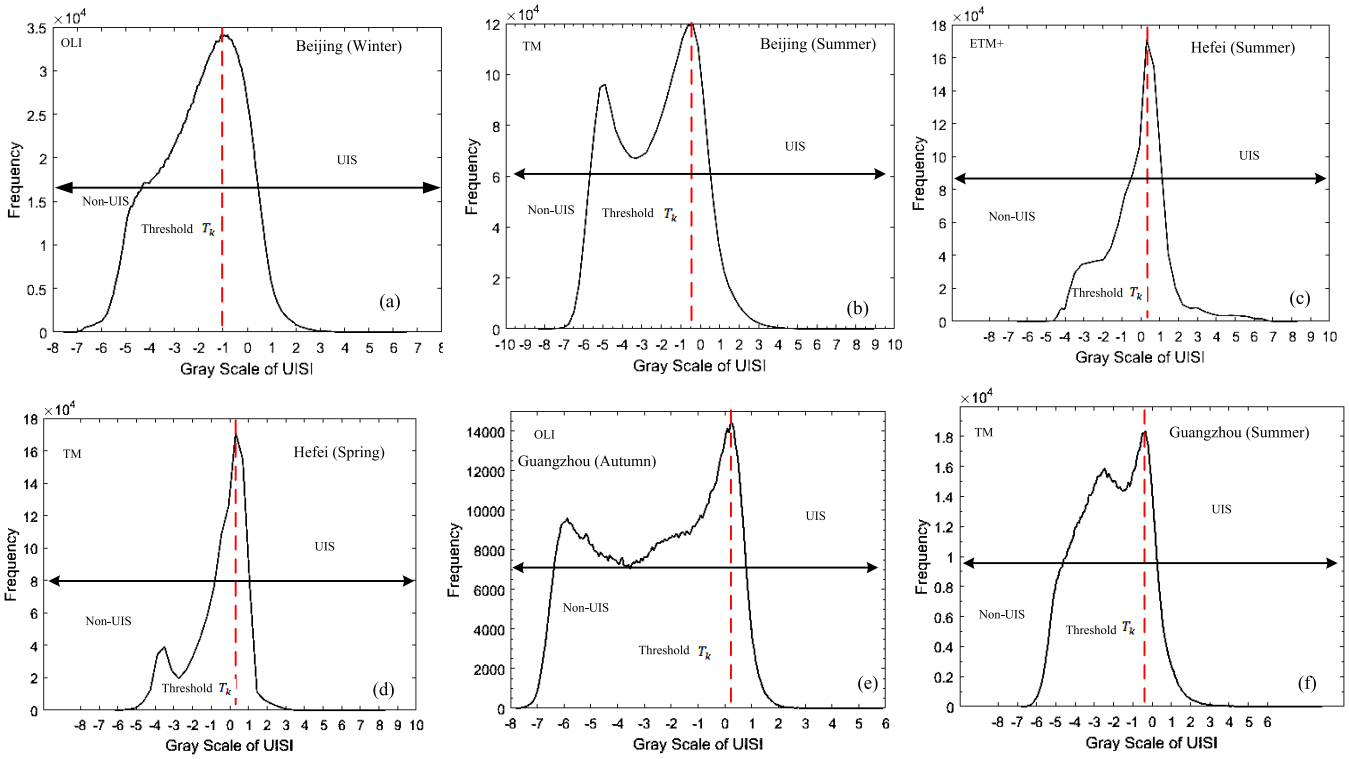


Fig. 7. Comparison of the threshold selection for the UISI index images derived from multitemporal Landsat images (OLI, ETM+, and TM) to discriminate UIS from non-UIS. The red dash lines mark the optimal threshold segmentation lines between UIS and non-UIS. The horizontal axis is the index value, and the vertical axis is the frequency of the index value occurrence in the index image.

Guangzhou, with the optimal thresholds T_k getting close to 0. In Fig. 7, the multitemporal Landsat images were selected, involving OLI, ETM+, and TM images and covering four seasons. The optimal thresholds of the UISI index images varied over time and space.

C. Accuracy Verification

The UIS extraction accuracy of the UISAT method was verified via a relative accuracy evaluation. The relative accuracy comparison is the accuracy verification method under the same reference object [58]. We selected a supervised learning classification method with higher accuracy, and its results were considered as a reference. Deep learning classification previously proved to have higher classification accuracy [36]. In this article, the dc-CDBN was applied for relative accuracy comparisons with UISAT. In particular, the dc-CDBN self-learned and classified the Landsat images into four ground object types, including UIS (high-albedo and low-albedo), vegetation, water, and bare soil. The accuracy of the dc-CDBN was proven in our previous works [37]. The pixel numbers of the four land cover types determined by the dc-CDBN were counted and compared with the threshold-based segmentation results from the UISI image to obtain the relative accuracy of the proposed model.

Landsat 8 OLI and five TM images collected during summer and early autumn were selected for the three cities to reduce the interference of natural bare soil. The producer's accuracy (PA) and user's accuracy (UA) were used to assessing the UIS extraction accuracy, for both dc-CDBN and UISI in the three cities (see Table III). The relative accuracy of UISI was estimated via a PA at 91.73%, 93.87%, and 92.84% and a UA at 91.51%, 93.35%, and 92.24% for Beijing, Hefei, and Guangzhou, respectively. Table III also compares the accuracy (PA and UA) between UISAT and other typical indices, including CBI, BCI, GA-UCAT, and G-MNDISI. We set $l = 0.5$ and $\rho = 1.438 \times 10^{-2}$ for CBI and G-MNDISI, respectively. Hefei segmentation results exhibited higher PA and UA values due to its lower urban spatial complexity properties, while the greater urban spatial density in Beijing resulted in lower accuracy. The UISAT model exhibited the highest extraction accuracy, which increased by 6.01% in comparison to the average accuracy measure rendered by other typical indices. However, the relative accuracy of the indices was related to the reference results from the dc-CDBN algorithm. Although the accuracy of the indices varied with different spatiotemporal Landsat images, the overall performance of the indices was not affected. Table III demonstrates that the proposed UISAT model was able to outperform other commonly used indices.

Fig. 8 presents the UISAT classification results of the downtown areas within the three cities. The ground objects were divided into four groups via the selected threshold (UIS, vegetation, water, and bare soil). Images were collected in the summer and early autumn seasons because the vegetation is in the leaf-on period at that time; thus, bare soil was minimal in

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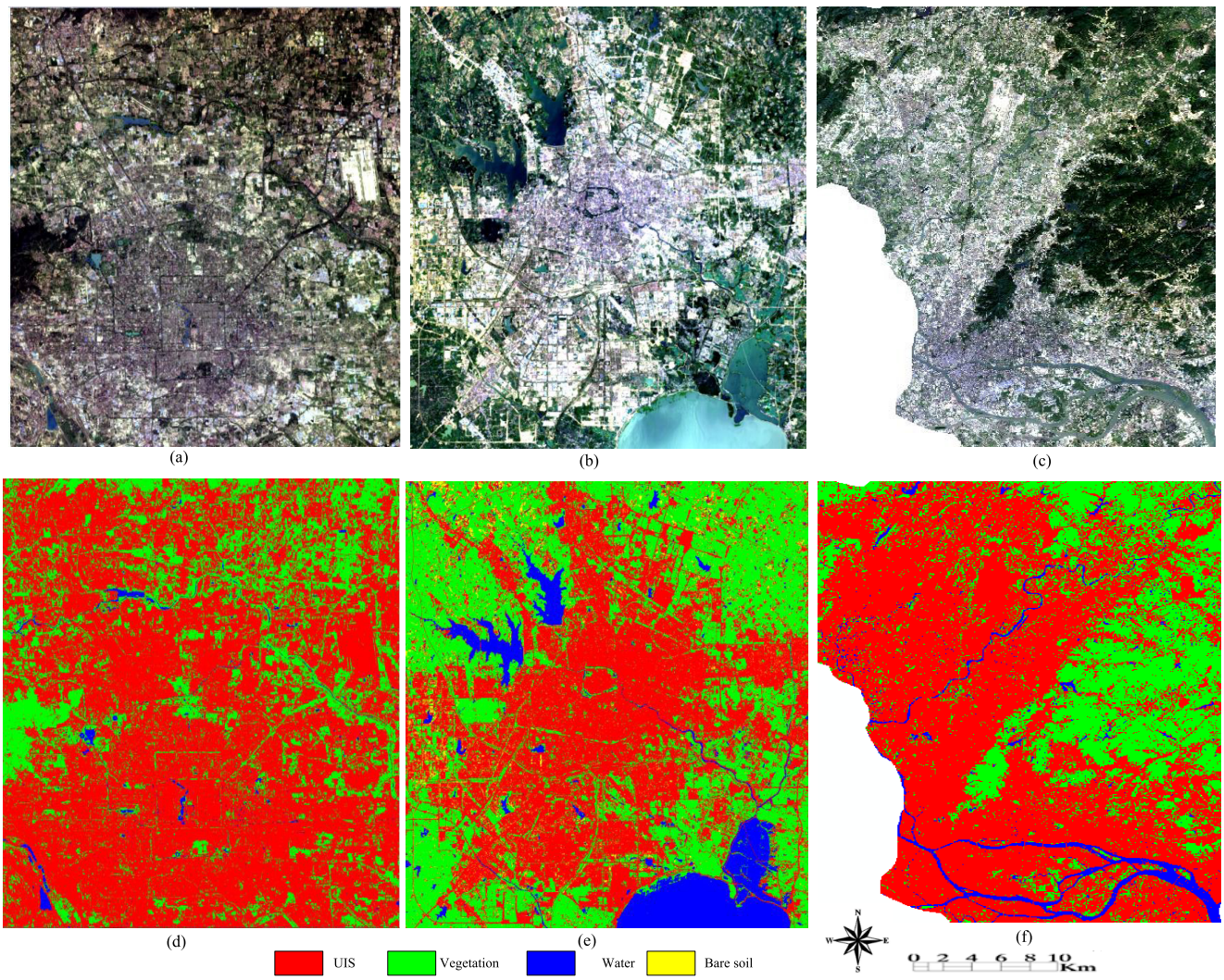


Fig. 8. Land cover classification results of the three main downtown areas by the dc-CDBN. (a)–(c) Original images of Beijing, Hefei, and Guangzhou, and the imaging time was August 19, 2014, October 11, 2015, and October 23, 2017, respectively. (d)–(f) Corresponding classification results by dc-CDBN.

TABLE III
UIS EXTRACTION ACCURACY COMPARISON BETWEEN THE UISAT MODEL AND THE TYPICAL UIS INDEX MODELS

City	dc-CDBN		CBI		BCI		GA-UCAT		G-MNDISI		UISAT	
	PA (%)	UA(%)	PA (%)	UA(%)	PA (%)	UA(%)	PA (%)	UA(%)	PA (%)	UA(%)	PA (%)	UA(%)
Beijing	94.31	93.92	85.71	84.28	85.42	85.17	87.73	86.55	87.84	86.31	91.73	91.51
Hefei	96.78	96.35	88.25	87.83	87.63	86.67	89.75	87.68	89.76	89.76	93.87	93.35
Guangzhou	95.83	95.12	86.44	85.61	86.94	85.73	88.34	86.51	90.62	89.51	92.84	92.24

the downtown areas. Higher building densities were observed in downtown Beijing and Guangzhou [see Fig. 8(a) and (c)] compared to downtown Hefei. Fig. 8 reveals the effects of building density on the extraction accuracy of UIS.

Fig. 9 compares the results between UISAT and dc-CDBN models. We selected representative parts of the three cities, including high-density building areas in downtown Hefei and Guangzhou, and a low-density building area north of Beijing. The blue ellipses denote areas of high similarity between dc-CDBN and UISAT results. The dc-CDBN method misclassified several old and densely built areas, such as those marked by black ellipses in Fig. 9(e) and (h). The UISAT tended to misclassify roads and streets covered by vegetation (trees on both sides of the road/street) in low-density building areas, such as non-UIS, including the marked positions of the black ellipses in Fig. 9(c) and (i). A few bare

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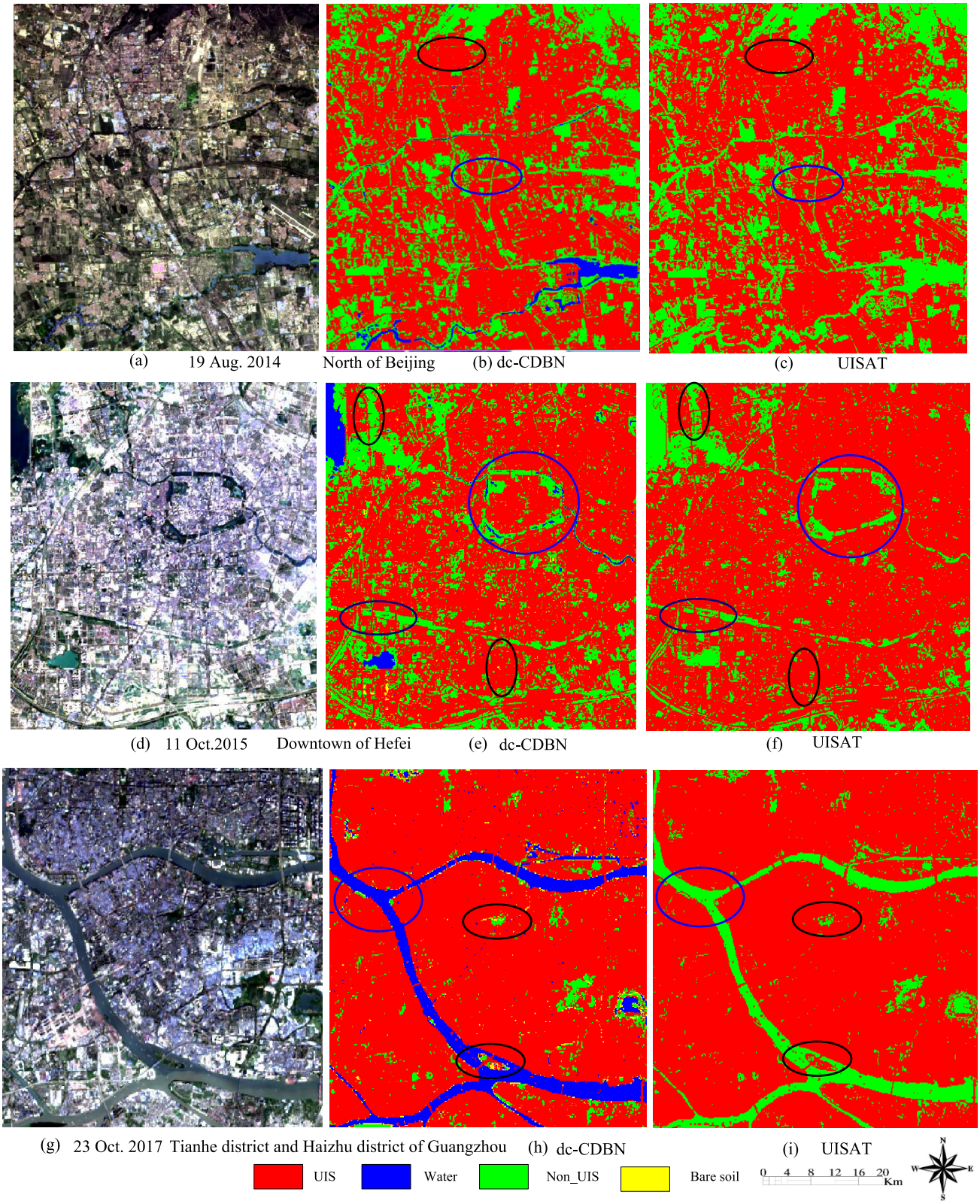


Fig. 9. Comparison of the UIS extraction results between dc-CDBN and UISAT in the three areas. (a), (d), and (g) Landsat images of a downtown part of Beijing, Hefei, and Guangzhou. (b), (e), and (h) Results from dc-CDBN. (c), (f), and (i) Results of UISAT. The blue ellipses show consistent results from both dc-CDBN and UISAT, and the black ellipses show inconsistent results between dc-CDBN and UISAT.

soil regions acted as a disturbance for UIS extraction in the north of Beijing, as well as in Hefei and Guangzhou [Fig. 9(f) and (i)].

Google Earth images were used to verify the accuracy of the UISAT model. A total of 300 random comparison points were generated in the Google Earth image within the

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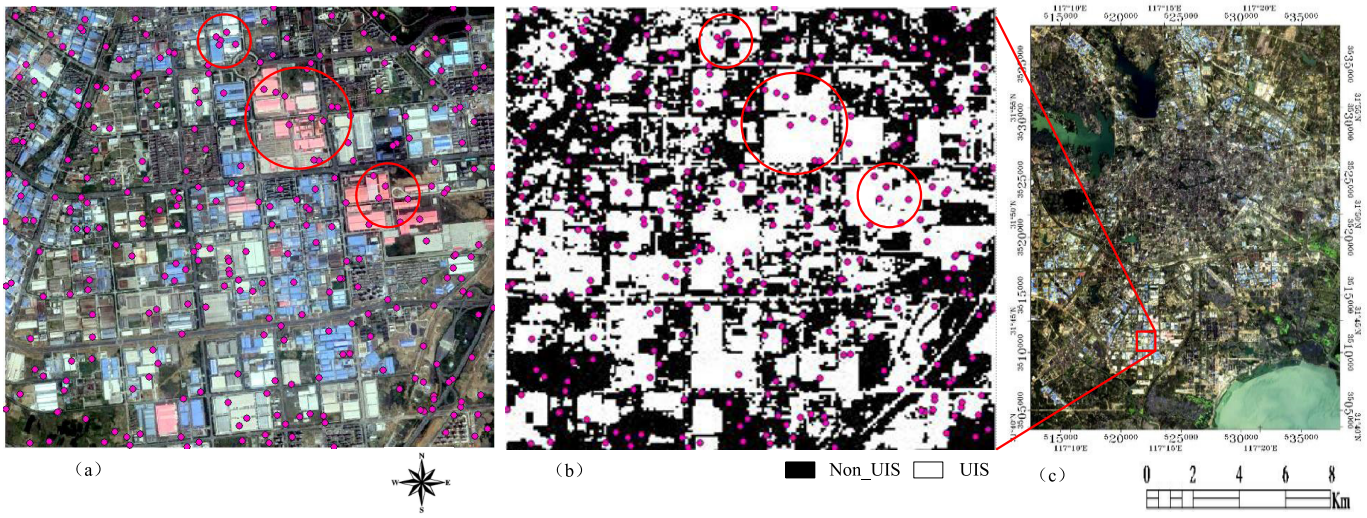


Fig. 10. Ground object types of the 300 random points in Google Earth image compared with UISI image: a part of Hefei downtown. The red circles highlight the comparison details between the Google Earth map and UISAT results. (a) Google Earth (1-May-2015), (b) UISI (11-Oct. 2015), (c) Hefei.

TABLE IV
ACCURACY EVALUATION OF UISAT MODEL DERIVED FROM 300 RANDOM POINTS IN PARTNERSHIP WITH GOOGLE EARTH IMAGE IN HEFEI, CHINA

Accuracy	Type of ground object	UISAT (number of pixels)				
		UIS	Non-UIS	Number of comparison pixels	PA (%)	97.5
Google Earth	UIS	271	7	278	UA (%)	97.8
image	Non-UIS	6	16	22	OA (%)	95.6
				300	Kappa	0.69

study area, and their true land cover types were visually interpreted as reference. At the same time, the mapped land cover types of the 300 points by UISAT were directly extracted from the optimal threshold segmented image. Comparing the two results (see Fig. 10 and Table IV—an example of part of Hefei), we could verify the effectiveness of the UISAT model. The red circles in Fig. 10 denote the comparison points at the same coordinates. The overall accuracy (OA), PA, and UA were computed as accuracy evaluation indicators.

The comparison of the relative accuracy of both UISAT and the typical indices is listed in Table III. We also compared the results of the proposed UISAT with the aforementioned typical indices on the previously selected downtown areas of Beijing, Hefei, and Guangzhou and took part of Guangzhou for example in Fig. 11. Fig. 11(a-1)–(a-5) compares the results of the UISAT and the typical indices in part of the Guangzhou area. The blue ellipses represent the subtle differences in the UIS index results. The UIS extraction results demonstrate the variation in performance of the typical indices for different spatiotemporal images, with the UISAT model exhibiting an improved performance. For example, BCI and GA-UCAT underestimated the UIS in areas with dense old buildings (blue ellipses), while several streets and old-low buildings were misclassified as non-UIS regions. In the Guangzhou area, roads lined with trees and sparse buildings were determined as non-UIS via CBI, BCI, and GA-UCAT models.

D. UISAT Performance Evaluation

The UISI image was divided into the foreground (UIS) and background pixels group using optimal threshold T_k, resulting in the foreground pixels group $\mathbb{R}_F$ and the background pixels group $\mathbb{R}_B$ for each UISI image. The ICC represents the similarity of the gray values between each pixel within a pixel group, with ICC values greater than 0.75 denoting a high correlation between pixels in terms of their gray value [37], [65]. Table V reports the ICC ($\mathbb{R}_F$ and $\mathbb{R}_B$) values of the UISI images for the three cities over four seasons. Abnormal values due to image noise were deleted before the ICC was calculated. The $\mathbb{R}_B$ correlations were less than those of $\mathbb{R}_F$. This can be attributed to the presence of vegetation, water, and bare soil pixels in the former group, with a wider gray value range and a larger number of pixels. The correlations of the foreground pixel group $\mathbb{R}_F$ were all greater than 0.75, denoting a highly accurate UIS extraction from the threshold T_k. The correlation between UIS and non-UIS pixel groups was determined as 0.325, representing a high segmentation degree between UIS and non-UIS regions.

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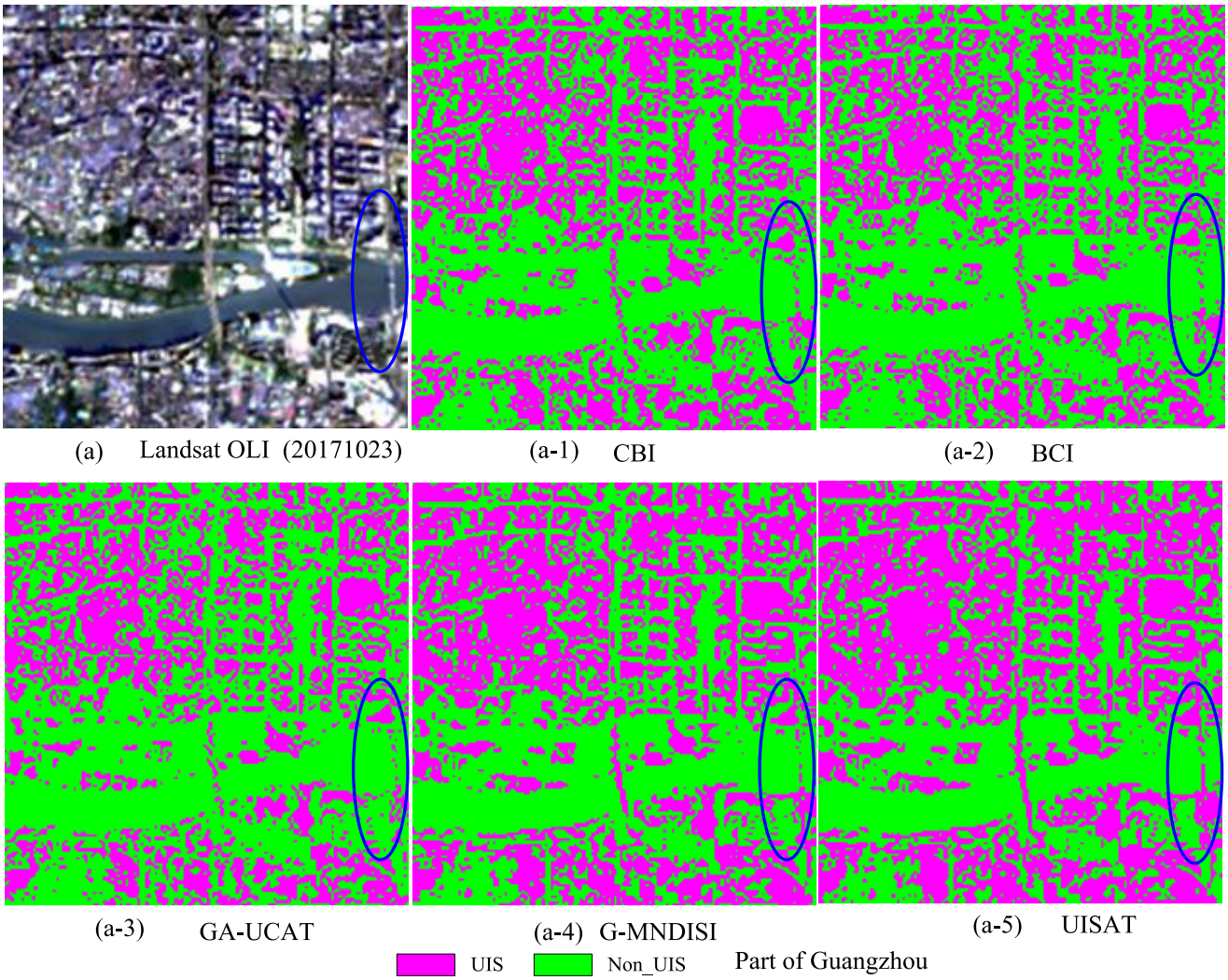


Fig. 11. Comparison of the UIS extraction between CBI, BCI, GA-UCAT, G-MNDISI, and UISAT in part of Guangzhou from Landsat OLI image (20171023). (a) Part of Guangzhou of the Landsat image. (a-1)–(a-5) Results from CBI, BCI, GA-UCAT, G-MNDISI, and UISAT, respectively. The blue ellipses are the comparison areas of the indices derived results.

V. Discussion

A. Bare Soil Disturbance and Season Selection

Bare soil is considered a principal interference factor for the detection and extraction of UIS due to its similar albedo to UIS [65]. In cities, bare soil can be divided into natural and artificial bare soil (exposed soil on construction sites). The spectral reflectance of natural bare soil varies with the vegetation cover ratio and season. In most parts of China, naturally, bare soil is generally covered with crops in late spring, summer, and early autumn and can easily be interpreted as non-UIS. However, artificial bare soil is compacted by human factors without vegetation coverage during the construction periods and exhibits a high spectral reflectance. Thus, artificial bare soil presents a greater level of interference compared with natural bare soil when mapping UIS.

Reducing the influence of artificial bare soil has been a great focus of attention amongst researchers [58]. A comparison of time series images can decrease the interference of bare soil [69]. Therefore, we selected multitemporal images that cover four seasons from three different sensors (OLI, ETM+, and TM) to increase the discrimination capability of land cover types (see Table I). However, a small number of artificial bare soil regions were still determined as UIS due to its spectral signature overlapping with UIS, thus influencing the OA of the model, but the influencing degree was limited. Basically, the area proportion of urban artificial bare soil within cities is relatively low; thus, UIS extraction accuracy is acceptable. In contrast to bare soil, vegetation and water pixels exhibited lower gray values and were distinct from UIS in the index images. The UISAT model reduced the differences derived from the spatiotemporal changes and sensor types through an automatic selection of the threshold in different images.

Remote sensing images collected in different seasons have different capabilities in presenting vegetation and then influencing the distribution of bare soil. In this article, we recommended imaging season selection when mapping UIS detection by index-dependent methods in Section III-A. However, the effects of spatial variation and building density should also be considered when using an index model to detect

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TABLE V
COMPARISON OF ICC BETWEEN UIS AND NON-UIS REGIONS IDENTIFIED FROM UISAT FOR THE THREE CITIES OVER FOUR SEASONS

City	Winter		Spring		Summer		Autumn	
	$\mathbb{R}_F$	$\mathbb{R}_B$	$\mathbb{R}_F$	$\mathbb{R}_B$	$\mathbb{R}_F$	$\mathbb{R}_B$	$\mathbb{R}_F$	$\mathbb{R}_B$
Beijing	0.845	0.609	0.836	0.625	0.861	0.642	0.854	0.618
Hefei	0.843	0.655	0.876	0.652	0.851	0.666	0.855	0.653
Guangzhou	0.861	0.632	0.859	0.635	0.861	0.622	0.865	0.548

UIS from Landsat images. For example, in northern China, the more suitable periods are late spring, summer, and early autumn (from May to October), while, in southern China (e.g., Guangzhou), the main interference derived from the rainy season from May to June and the suitable imaging season will shift from November to January of the following year. The UIS detection accuracy of dense building areas was higher than that of sparse building areas also due to the difference in vegetation cover ratio. Therefore, the threshold adjustment should be time- and space-specific, and sensor-dependent.

SAR and LiDAR data were used to detect UIS changes and mapping due to their rich textures and 3-D geometric information, which can complement the drawback (e.g., environmental and weather disturbances) of Landsat images of the environmental and weather disturbances. Using SAR and LiDAR data can reduce the interference of bare soil and enhance the identification of UIS from bare soil. In addition, the UIS accuracy detection was previously shown to be improved by approximately 3.5% via data fusion between SAR and LiDAR [41]. This provides a fusion approach for improving the detection accuracy of UIS in a specific urban region. However, for long-time continuous UIS detection on a regional scale, the availability of SAR and LiDAR data remains an ongoing challenge.

B. Automatic Threshold Selection

Automatic threshold selection can avoid manual frequent fine-tuning and reselection for different spatiotemporal images. UIS detection can be divided into three steps using the UIS indices: spectral bands selection, bands' calculation, and threshold selection. A discerning threshold is a key to UIS detection by UIS indices because it directly determines UIS detection accuracy. However, the automatic threshold selection method sometimes underestimates (high-threshold) and overestimates (low-threshold) UIS in urban areas. A low threshold may occur in old and dense urban areas due to the lower spectral reflectivity, and a high threshold may occur in the rural-urban fringe areas due to the higher spectral reflectance of colored steel tiles. To avoid the high and low thresholds of the UISAT model, we adopted the dynamic inertia weight to adjust the algorithm convergence and achieve a global optimal threshold.

The automatic threshold selection method of the UISAT had some advantages over the genetic- and Gaussian-based methods. These advantages could be summarized as follows: 1) the UISAT model, compared with the genetic-based method, avoids inaccuracies of coding, uses fewer parameters, and achieves a higher accuracy [58]; 2) the UISAT model uses a single information-sharing mechanism and obtains a better search speed than the other two methods; 3) the image entropy method is applied for the fitness function of the improved PSO-based method, which ensures an optimal threshold by considering the maximum image entropy criterion [64], [65]; and 4)multidimensional efficacy evaluation strategy for UISAT to ensure its correctness and reliability. However, the UISAT model has not been tested by using other images [e.g., RapidEye and Systeme Probatoire d'Observation de la Terre (SPOT)]. This will be our recent mission to test their effectiveness.

The accuracy of the automatic threshold selection method also depends on the capability of the proposed UIS index to express or depict different land cover types. The current UIS index transforms original Landsat images into index images with enhanced UIS gray scale and subdued non-UIS gray scale. The enhanced discrimination is helpful to improve the threshold using the automatic selection method. Therefore, the UIS index development or selection is a necessary precondition for automatic threshold search. Moreover, automatic threshold search fully considers the effects of different locations, different seasons, and sensors on the distribution of pixel values in the UIS index image. Our experiment showed that the influence of geospatial variation on threshold selection was greater than that of seasonal changes (see Fig. 7).

In addition, the accuracy of the UISAT model is affected by the value of φ in (4). The value of φ is recommended to be in the range of 1–3 from the spatiotemporal Landsat images experiments. The gray values of some UIS pixels are more than 10 when $\varphi > 3$ and less than 0.01 when $\varphi < 1$, leading to calculation difficulty. The empirical value of φ is selected for 2 in this article to reduce accuracy deviation and computational complexity. The value of φ can be readjusted to adapt to different spatiotemporal Landsat images.

C. Verification and Evaluation

The relative accuracy verification was widely applied in the performance verification and evaluation of UIS indices, such as SVM-based [32], maximum likelihood-based [57], and combined SVM and linear SMA [70]. Compared with

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these strategies, dc-CDBN has less human intervention and less established sample training and testing with self-learning and provides hierarchical convolution architecture for spatial feature detection. However, the dc-CDBN algorithm has omission errors and misclassifications, such as other deep learning algorithms, due to the influence of the self-constructing sample quantity and quality. Some samples contain a lot of mixed pixels that are easy to confuse, particularly in areas with complex and heterogeneous land cover types. Therefore, ground truths are essential for validation purposes. The 2.39-m spatial resolution Google Earth images were used for performance comparison as ground objects validation in the current work, which was a necessary supplement for the quantitative validation of dc-CDBN, and this enhanced the credibility of accuracy verification.

The spectral discrimination index (SDI) was adopted in the traditional performance evaluation of the UIS index model by Sun *et al.* [45] and Sun *et al.* [58]. The SDI was defined as the degree of segmentation between the UIS and non-UIS in spectral histograms. A higher SDI represents better threshold segmentation for index images. Compared with SDI, the ICC used in this analysis can evaluate not only the degree of segmentation between UIS and non-UIS but also the intraclass correlation of UIS and non-UIS pixels groups. The intraclass correlation represents the similarity of the pixel gray scale within a group, and the extra-class correlation represents the segmentation degree of different pixel groups. Therefore, the ICC provides a credible and comprehensive evaluation of the proposed model.

D. Research Value of the UIS Detection

Reliable UIS detection is a crucial contribution to urban ecological remote sensing research. The mapped UIS current state and dynamic trend help develop a reasonable spatial layout of urban elements when formulating a sustainable urban management strategy by adequately considering urban ecological environment quality and living standards. In addition, the results of UIS mapping are helpful for a comprehensive urban development study including transportation, environment, population, and socioeconomic policies to facilitate governmental decision-making.

VI. Conclusion

In this article, we first attempted to propose the UISAT model for UIS detection from multitemporal Landsat images. The proposed UISAT model consisted of a novel UIS index (UISI), an improved PSO-based automatic threshold selection method, and a combined evaluation strategy.

1) The UISI generated UIS index images through the multiband operation with an enhanced UIS pixel gray scale and improved the identification between UIS and non-UIS.

2) The PSO-based automatic threshold selection reduced the manual interference for threshold fine-tuning and reselection and considered the impacts of different imaging locations, seasons, and sensors on the capability of UIS index image in expressing different land cover types. Automatic threshold selection improved the OA of UIS extraction from the Landsat images.

3) A combined evaluation strategy ensured the correctness and efficiency of the UISAT model by using both qualitative and quantitative performance evaluation methods. Overall, the UISAT model had a better performance in mapping UIS compared with other typical UIS indices and higher computational efficiency in comparison to a deep learning algorithm, e.g., dc-CDBN.

Further improvements to the proposed UISAT model may be needed when using other remote sensing images to map UIS. Multisource data fusion can improve UIS detection accuracies, such as night-light data, SAR, and LiDAR data. Combining these data with Landsat data could overcome the shortcomings of optical data to improve the UIS detection accuracy.

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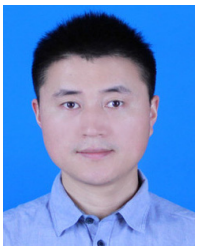
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